

		$\Rightarrow W = \frac{1}{3}Fx \Rightarrow F = 3kx$
6	B	Work done in stretching the material from 0 to T is area under tensile load against extension graph OPQ. Area under QRS is the energy recovered from the elastic potential energy originally stored in the specimen. Net work done on the specimen is area OPRS.
7	C	Let the force provided by the engine be F